

SANO RODRIGUE

+250 783 242 318 | r.sano@alustudent.com | Kigali, Rwanda
[linkedin.com/in/rodrigue-sano-ab3849331](https://www.linkedin.com/in/rodrigue-sano-ab3849331) | github.com/SanoRod00 |
<https://sano-rodrigue-official-portfolio.vercel.app/>

PROFESSIONAL SUMMARY

Builder and problem-solver with experience in hardware, web development, and AI tools. Co-led PolyVoice , a real-time platform helping deaf and hearing people communicate from idea to working product at CodeXtrem Hackathon 2025. Brings both technical skills and a focus on real user needs to build digital products that work for African markets.

CORE COMPETENCIES

Product & Strategy:

- User Story Writing
- Product Roadmapping
- User Journey Mapping
- Agile Methodologies
- Stakeholder Communication
- Feature Prioritisation

Technical:

JavaScript · Python · C/C++ · HTML/CSS · Node.js · Express.js · React · Socket.IO · REST APIs

AI & Automation:

Prompt Engineering · OpenAI APIs · Claude AI · AI Workflow Design · AI-Augmented Productivity

Hardware & Embedded:

Arduino · Raspberry Pi · Circuit Design (EasyEDA) · PLC/Ladder Logic · Tinkercad

Tools:

Git · GitHub · AWS · Linux · Figma · Draw.io · VS Code

WORK EXPERIENCE

Web Developer | Uburiza Solution | Kigali, Rwanda (Remote) | Jan 2026 – Present

- Developed and maintained full-stack web applications using React and Node.js for real clients

- Collaborated in agile remote team, independently managing task delivery and sprint progress
- Implemented REST API integrations, reducing manual data-handling workflows across client projects

Computer System & Architecture Instructor | Bridge TVET School | Kigali, Rwanda | Mar 2026 – May 2026

- Designed and delivered complete Level 5 ICT curriculum (ICTCSA5001) for 20+ students, built from scratch with no prior template
- Created three-stage pedagogical workflow (Tinkercad simulation → Draw.io documentation → real hardware), cutting student onboarding time significantly
- Produced 10+ session plans, Kahoot assessments, and practical exam guides for Arduino Mega 2560 and Raspberry Pi 3B projects
- Introduced industrial PLC ladder logic (PLC Fiddle, TIA Portal) as a module, bridging classroom theory to real-world automation

PROJECTS

PolyVoice: Accessible Communication Platform | CodeXtrem Hackathon | 2025 | Role: Product Lead & Co-Founder

- Identified critical accessibility gap: deaf and hearing users cannot communicate naturally over calls; scoped product, defined user personas, and prioritised features under 48-hour constraint
- Resolved team misalignment by mapping complete deaf-user journey on whiteboard, reframing debate from "my feature vs yours" to "what does the user need"
- Co-shipped real-time platform with live audio captions, sign language detection, and TTS using Node.js, Socket.IO, and OpenAI Whisper/TTS APIs

Beat Marketplace App: African Music Economy | Product in Development | 2025

- Conceptualising dual-sided marketplace for music producers and artists across Rwanda, Nigeria, Kenya, and Ghana; conducting user research and market sizing
- Designing stablecoin payment flows (USDC/USDT) and NFT-based licensing on Polygon to solve beat piracy at scale
- Mapping full user journeys for both producer (seller) and artist (buyer) sides of the platform

PhishGuard: Anti-Phishing Security Analyser | ALU Project | 2025

- Built and deployed full-stack security tool integrating 3 external APIs (VirusTotal, Disify, Abstract Email Validation) to detect phishing threats in real time
- Deployed across 2 AWS servers (Web01/Web02) with Nginx round-robin load balancing and SSH ProxyJump routing

Smart Irrigation System | ALU Robotics Hackathon | 2025

- Built automated sensor-driven irrigation system in under 24 hours; integrated soil moisture sensors with software logic to trigger irrigation in real time

Inzu-Core: Modular Incremental Housing | ALU Group Project | 2025

- Co-designed human-centred modular housing concept for informal workers in Kigali; created detailed user persona (Bosco, 28, moto-taxi driver) grounding the design in real lived experience
- Recognised by panel as "the first time students talked about a real person, not just a problem"

EDUCATION

- BSc Software Engineering | African Leadership University | Kigali, Rwanda | May 2025 – 2028 (Expected)

CERTIFICATIONS & TRAINING

- ALX AI Career Essentials: AI Augmented Professional Development | ALX | Nov 2024
- ALX Pathway Foundations: 8-Month Tech & Career Skills Programme | ALX | Aug 2025
- Full Stack Web Development Level 2: JavaScript & ReactJs | Tech Tonic Africa | 2025

LANGUAGES

- English: Professional Working Proficiency
- Kinyarwanda: Native